hexdiff(1) — Unix Reference Manual & Specification

TRANSFORMING HEXDIFF.C INTO A ROBUST UNIX UTILITY (READING C SPECIFICATION)

NAME

hexdiff — binary file comparison utility with side-by-side hex and ASCII dump output

SYNOPSIS

```
hexdiff [-bwt] [-c cols] [-s skip] file1 file2
hexdiff -h | --help
hexdiff -v | --version
```

DESCRIPTION

The **hexdiff** utility compares two binary or text files byte-by-byte and outputs a formatted, side-by-side hexadecimal and ASCII dump highlighting differences. Designed to adhere strictly to traditional Unix command-line conventions, it bridges the gap between raw binary inspection and standard POSIX diff utilities.

When either *file1* or *file2* is specified as `-`, **hexdiff** reads from standard input (stdin), allowing seamless integration into Unix pipeline filters.

OPTIONS

Option	Description
`-b, --brief`	Output only a summary indicating whether files differ (similar to `cmp -s`), returning standard exit codes.
`-w, --wide`	Expand line width to display 32 bytes per row instead of the default 16 bytes.
`-t, --terse`	Suppress ASCII column output, showing hex offset and values only.
`-c, --cols N`	Set custom column width *N* bytes per row (must be a positive multiple of 4).
`-s, --skip BYTES`	Skip the first *BYTES* of both input files before commencing comparison.
`-h, --help`	Display usage summary and exit with status 0.
`-v, --version`	Output version information and license details to standard output.

EXIT STATUS

The **hexdiff** utility exits with one of the following values:

- **0** — Files are identical (no differences found).
- **1** — Files differ (mismatches detected in payload or length).
- **>1** — An error occurred (file not found, permission denied, memory allocation failure, or invalid argument).

UNIX INTEGRATION & PIPELINE DESIGN

> **Design Principle:** By incorporating `getopt_long`, standard stream redirection via `-`, and POSIX-compliant exit codes, `hexdiff.c` transitions from a simple diagnostic script into a standard-compliant Unix filter component.

Standard Pipeline Usage

```
# Compare two live memory dumps with brief output
hexdiff --brief /var/log/core.0 /var/log/core.1

# Pipe compressed stream output directly into hexdiff
zcat firmware_v1.bin.gz | hexdiff - firmware_v2.bin

# Integrate with xargs for batch binary audit
find build/ -name "*.bin" | xargs -I {} hexdiff reference.bin {}
```

IMPLEMENTATION ARCHITECTURE

The core engine handles buffered memory mapping or sequential block reads to maintain minimal memory overhead (*O(1)* space complexity beyond block buffers). Unprintable characters in the ASCII column are safely sanitized into dot (.) placeholders, preserving terminal integrity.